

Module 1.1 – Hard Rules

Sets

I = set of invigilators, J = set of exam shifts, C = set of campuses.

Predicates

$Assign(i, j)$: invigilator i is assigned to shift j .

$Busy(i, j)$: invigilator i is busy during shift j .

$Overlap(j, k)$: shifts j and k overlap in time.

Hard Rules

R1 – No double-booking

$$\forall i \in I, \forall j, k \in J : \quad Overlap(j, k) \rightarrow \neg(Assign(i, j) \wedge Assign(i, k)).$$

Equivalent clause form:

$$\forall i \in I, \forall j, k \in J : \quad Overlap(j, k) \rightarrow (\neg Assign(i, j) \vee \neg Assign(i, k)).$$

R2 – Availability

$$\forall i \in I, \forall j \in J : \quad Busy(i, j) \rightarrow \neg Assign(i, j).$$

R3 – Exact staffing capacity

$$\forall j \in J : \quad \sum_{i \in I} Assign(i, j) = cap(j),$$

where $cap(j)$ is the required number of invigilators for shift j .